CS & IT ENGINEERING

DIGITAL LOGIC

Combinational Circuit



Lecture No. 11



By- CHANDAN SIR

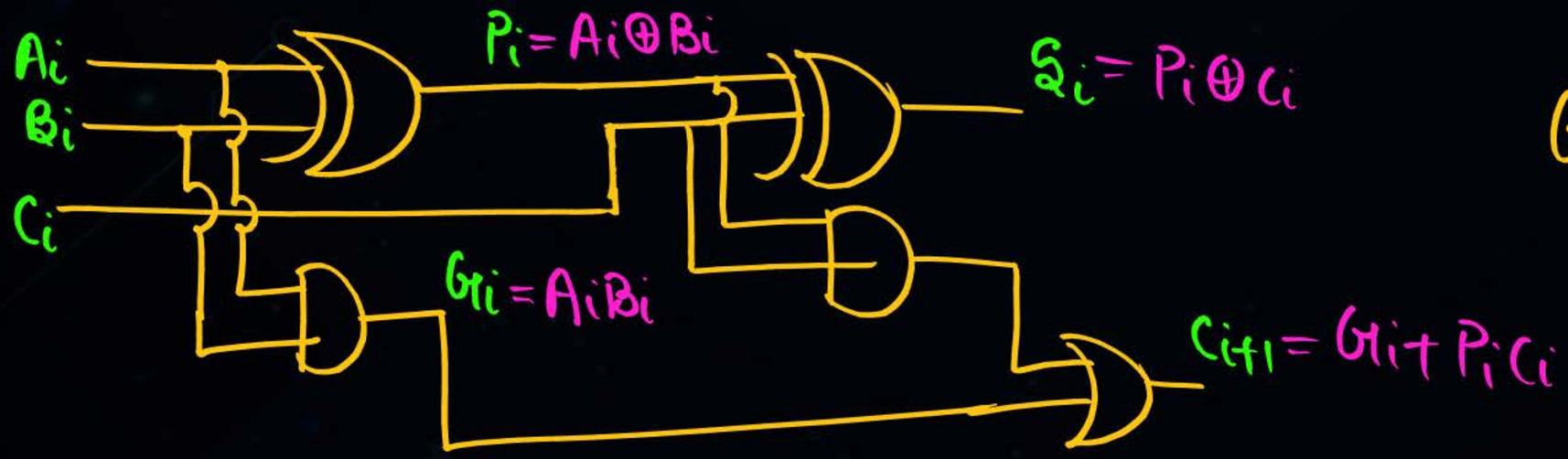


01 Question Practice

02 Discussion

Look Ahead Carry Adder {LACA}

↳ It is the fastest adder among all the adder.



Pi = Carry propagating term

Gi = Carry generating term

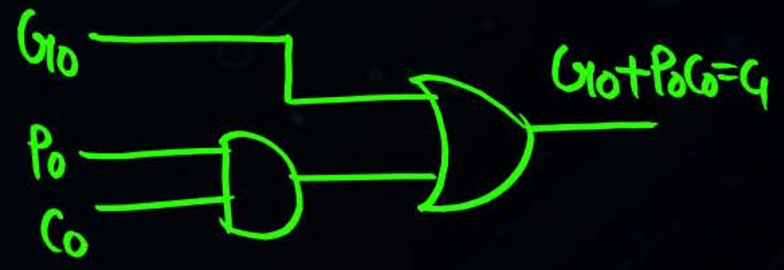
$P_i = A_i \oplus B_i$ $\quad G_i = A_i B_i$ $\quad S_i = P_i \oplus C_i$ $\quad C_{i+1} = G_i + P_i C_i$

$P_0 = A_0 \oplus B_0$ $\quad G_0 = A_0 B_0$ $\quad S_0 = P_0 \oplus C_0$

$P_1 = A_1 \oplus B_1$ $\quad G_1 = A_1 B_1$ $\quad S_1 = P_1 \oplus C_1$

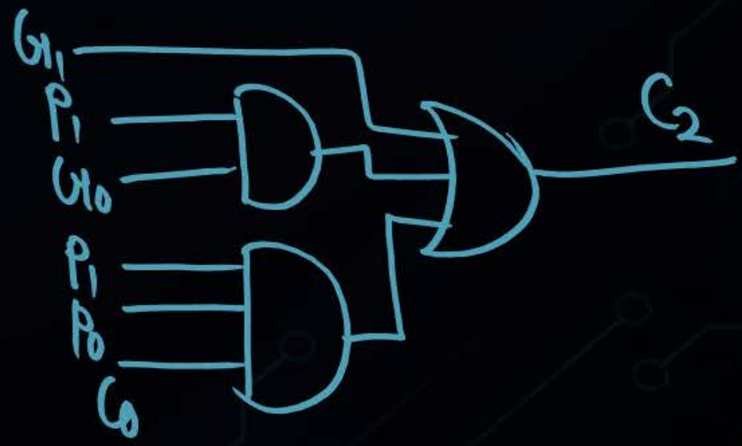
$P_2 = A_2 \oplus B_2$ $\quad G_2 = A_2 B_2$ $\quad S_2 = P_2 \oplus C_2$

$$C_1 = G_0 + P_0 C_0$$



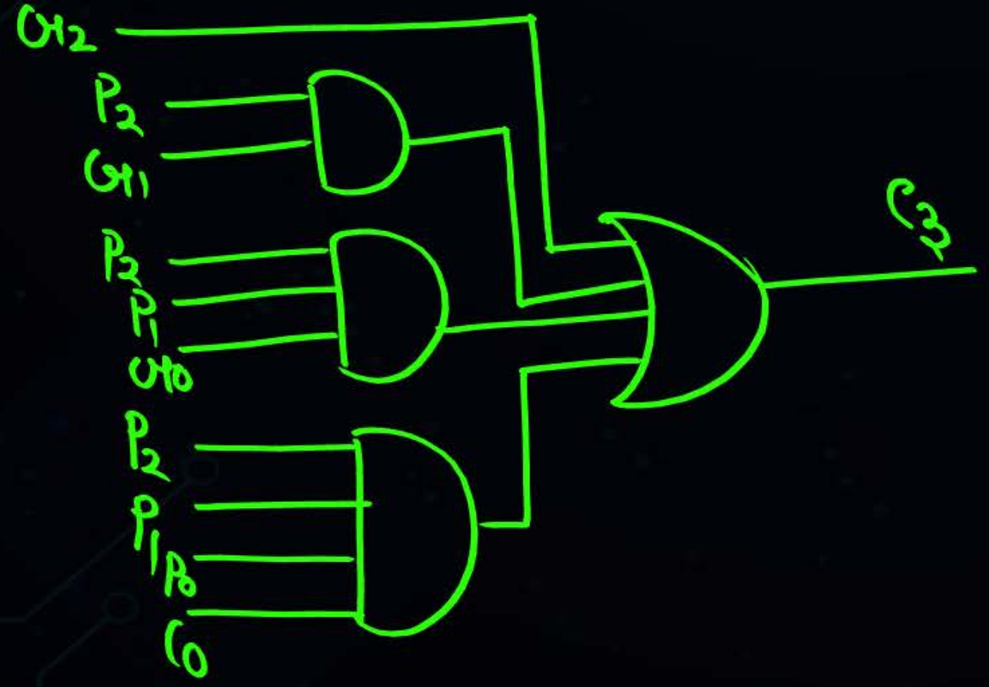
$$C_2 = G_1 + P_1 C_1 = G_1 + P_1 \{G_0 + P_0 C_0\}$$

$$C_2 = G_1 + P_1 G_0 + P_1 P_0 C_0$$



$$C_3 = G_2 + P_2 C_2$$

$$C_3 = G_2 + P_2 G_1 + P_2 P_1 G_0 + P_2 P_1 P_0 C_0$$



✯ Total Delay

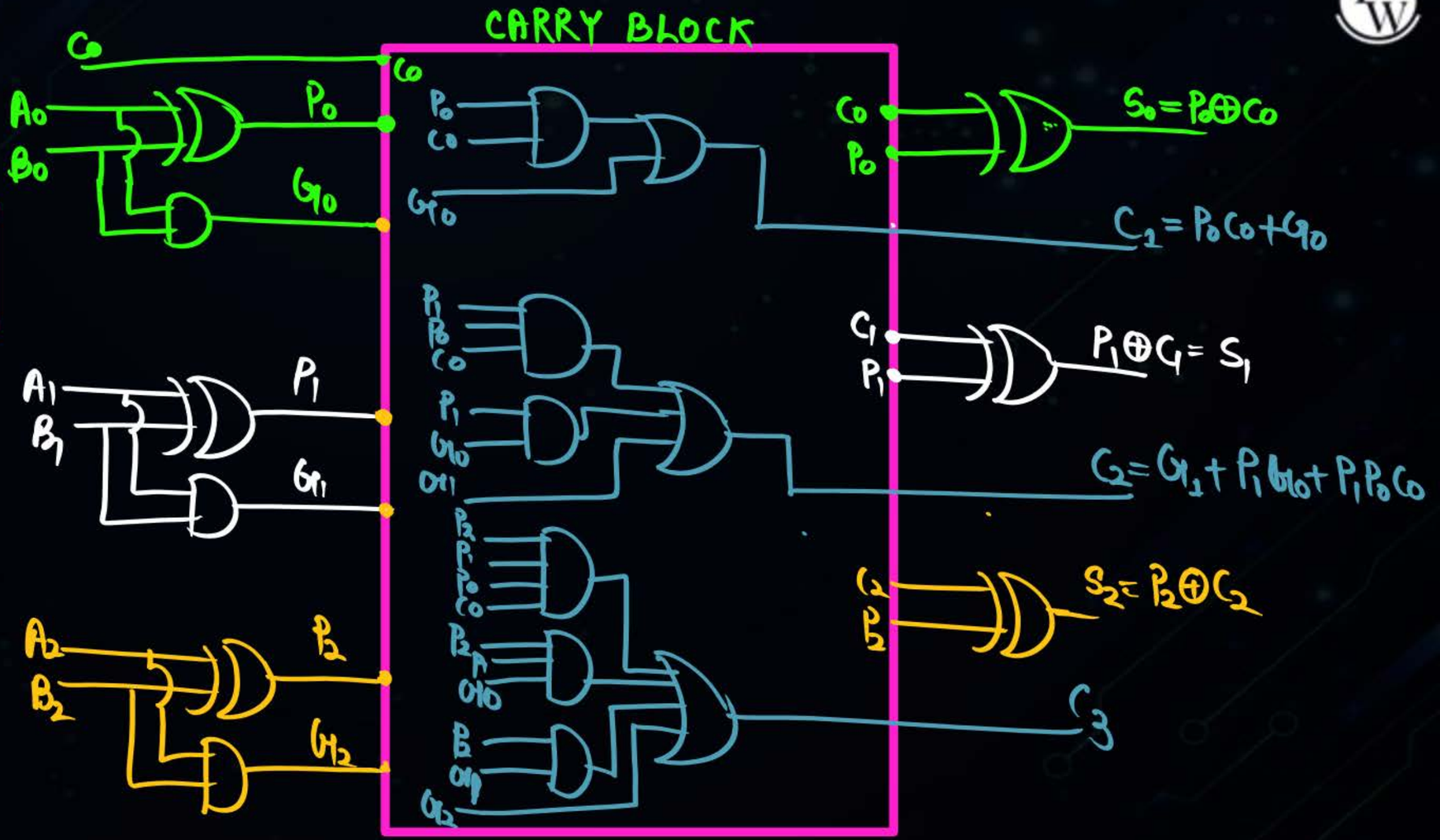
$$T = T_{XOR} + T_{AND} + T_{OR} + T_{X\text{-}OR}$$

$$T_{XOR} = T_{AND} = T_{OR} = \tau_{pd}$$

$$T = 4\tau_{pd}$$

✯ Carry Block Delay

$$T = T_{AND} + T_{OR}$$



Carry Block

Number of AND GATE $= \frac{n(n+1)}{2}$

Number of OR GATE $= n$

3bit LACA

AND $= \frac{3(4)}{2} = 6$

OR $= 3$

MULTIPLIER :→

2 bits Multiplier

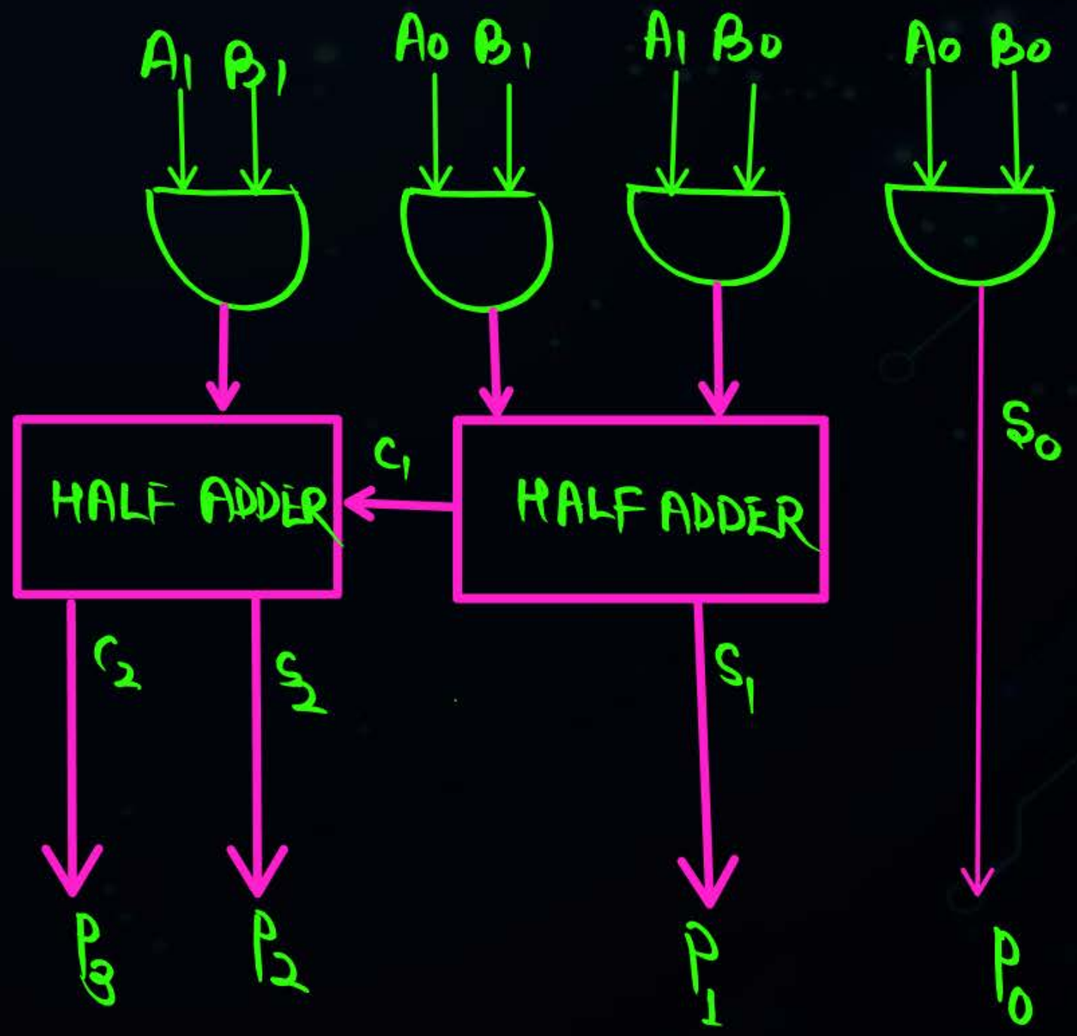
A_1 A_0

B_1 B_0

C_1	A_1B_0	A_0B_0
A_1B_1	A_0B_1	—

C_2 S_2 S_1 S_0

P_3 P_2 P_1 P_0



Q Design a Multiplier

$A_2\ A_1\ A_0$

$\times\ B_1\ B_0$

Question **(MCQ)**

#Q. How many half adders, will be required to add two *k* bit numbers?

A. 2k + 1

B. 2k – 1

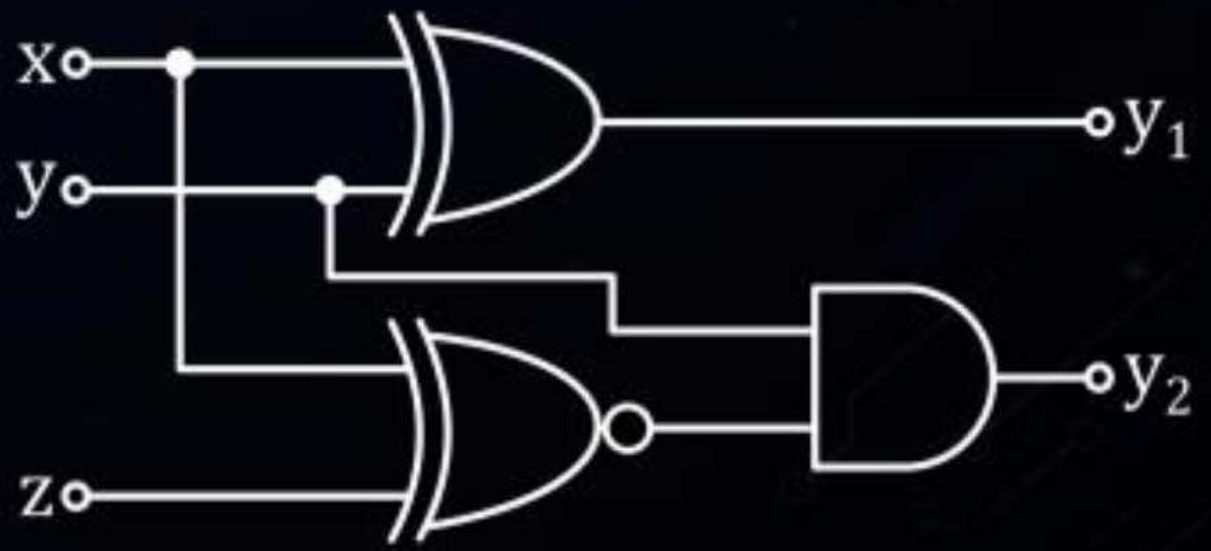
C. 2k

D. 2(k + 1)

$(2k-1)HA+(k-1)OR$

Question **(MCQ)**

#Q. The circuit shown below, is a controlled half adder/half subtractor. The inputs to half adder/half subtractor are x and y while z is a control. The outputs are y_1 and y_2.
Then the circuit behaves as



A Half adder for z = 0

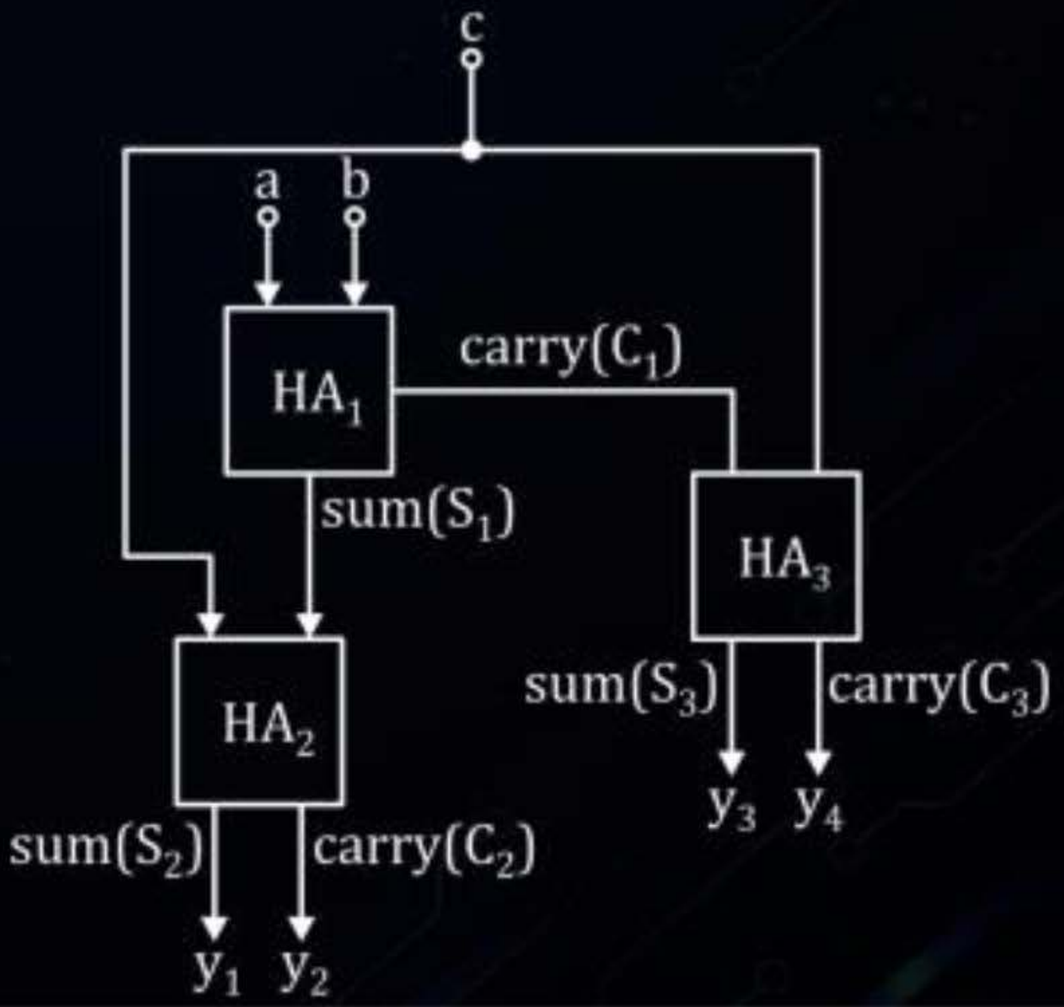
B Half subtractor for z = 1

C Half adder for z = 1 and half subtractor for z = 0

D Half adder regardless of weather z = 0 or z = 1 due to design defect

Question **(MCQ)**

#Q. Three half adders HA_1, HA_2 and HA_3 are inter-coupled as shown below. The four output functions y_1, y_2, y_3 and y_4 are expressed in terms of inputs a, b and c. Which one of the following output expressions is correct?



A $y_1 = (a \oplus b)\, c$

B $y_2 = (a \oplus b) \oplus c$

C $y_3 = ab \oplus c$

D $y_4 = a\,(b \oplus c)$

Question **(MCQ)**

#Q. The design of a combinational logic circuit with three inputs x, y, z and three outputs A, B, C is attempted. The constraint is that designer has only HA, HS, FA and FS units only in his inventory.
When the binary input is 0, 1, 2 or 3 the binary output is same as input and when binary input is 4, 5, 6 or 7 the binary output is 2 less than binary input. What completes the design?

A One FA and one HS

B One HA and one HS

C One HA only

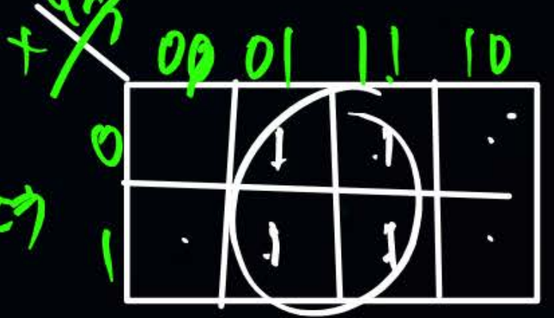
D One FA only

X	Y	Z	A	B	C
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	0	1	0
1	0	1	0	1	1
1	1	0	1	0	0
1	1	1	1	0	1

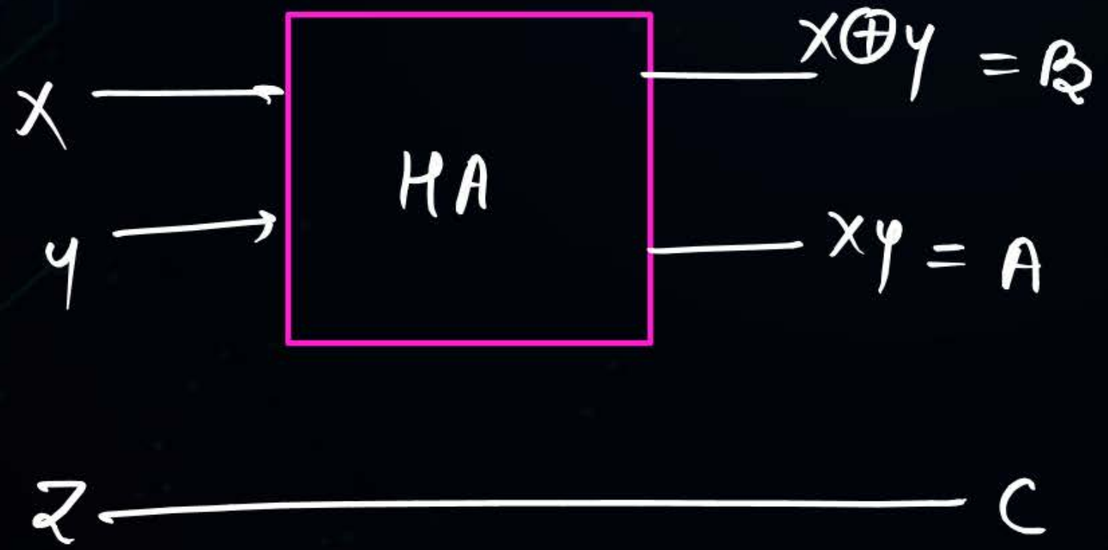
$A = xy\bar{z} + xyz = xy(\bar{z}+z) = xy$



$B = x\bar{y} + \bar{x}y$
$= x \oplus y$



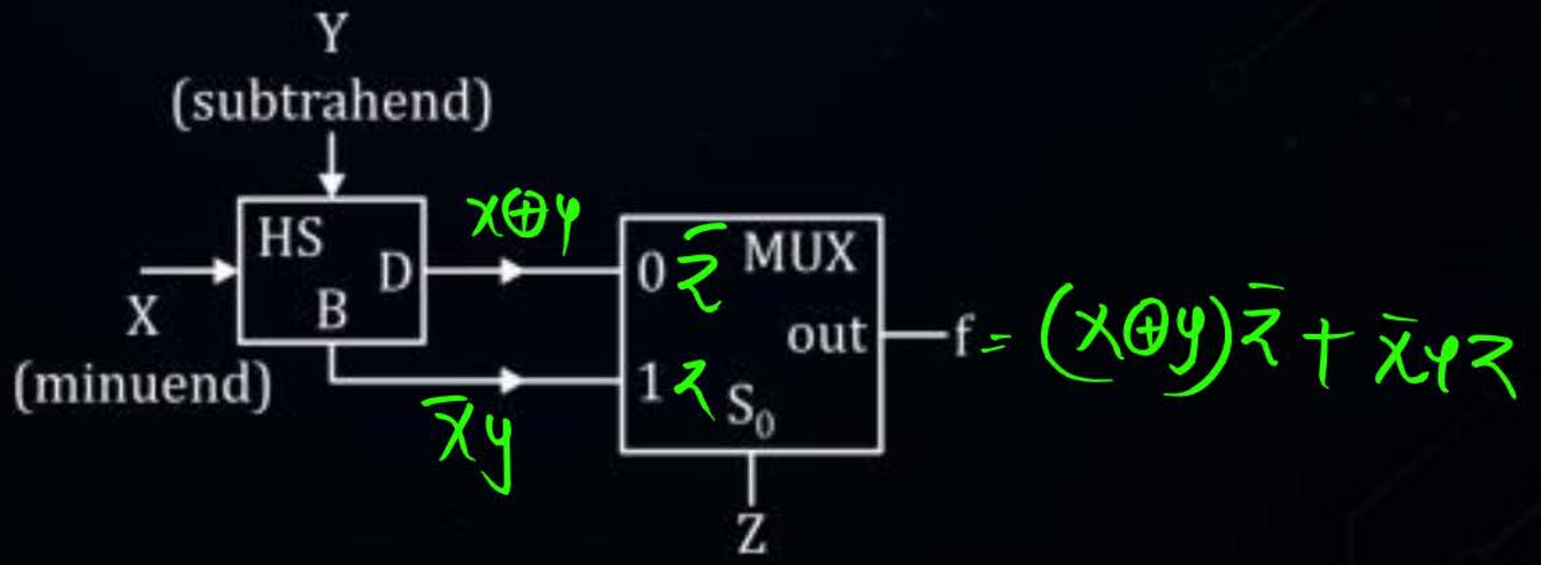
2


X
Y
HA
X⊕Y = B
XY = A
Z
C

Question **(MCQ)**

#Q. A half subtractor (HS) and 2 × 1 MUX are inter-connected as demonstrated below. What is NOT correct about this circuit?

HS



A For Z = 0, f = 1 indicates that the minuend and the subtrahend bits are different

B For Z = 0, f = 0 indicates that the minuend and the subtrahend bits are same, that is, X = Y = 0 or X = Y = 1

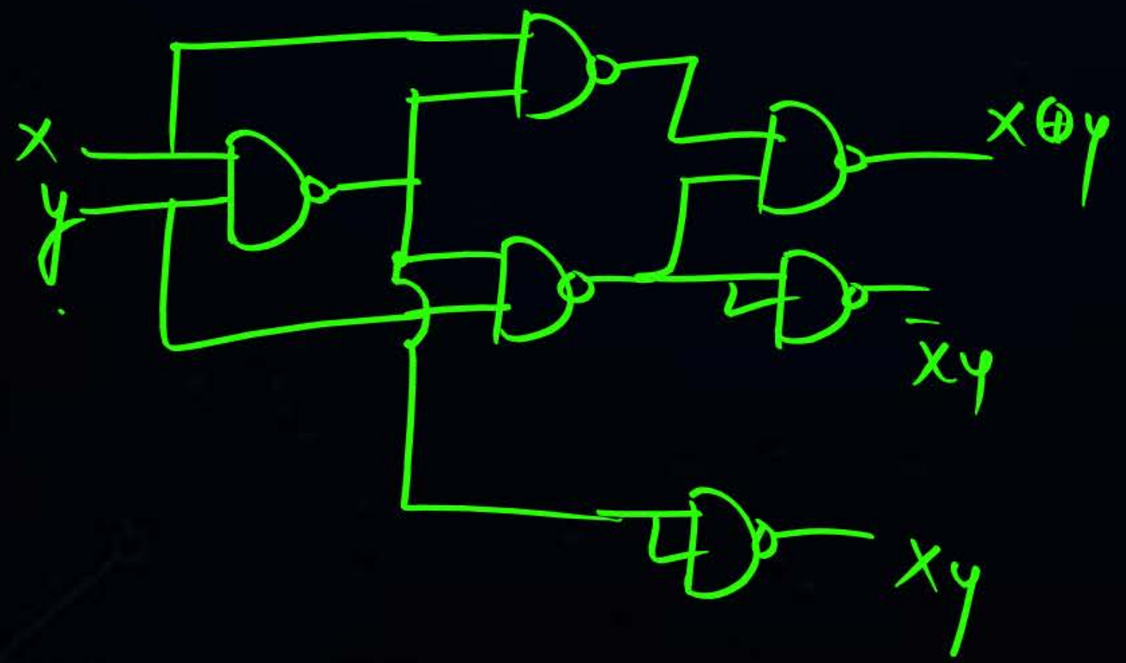
$(x \oplus y)\bar{z} + \bar{x}yz$

C For Z = 1, f = 1 indicates that X < Y

D For Z = 1, f = 0 indicates that subtrahend bit is definitely 0.

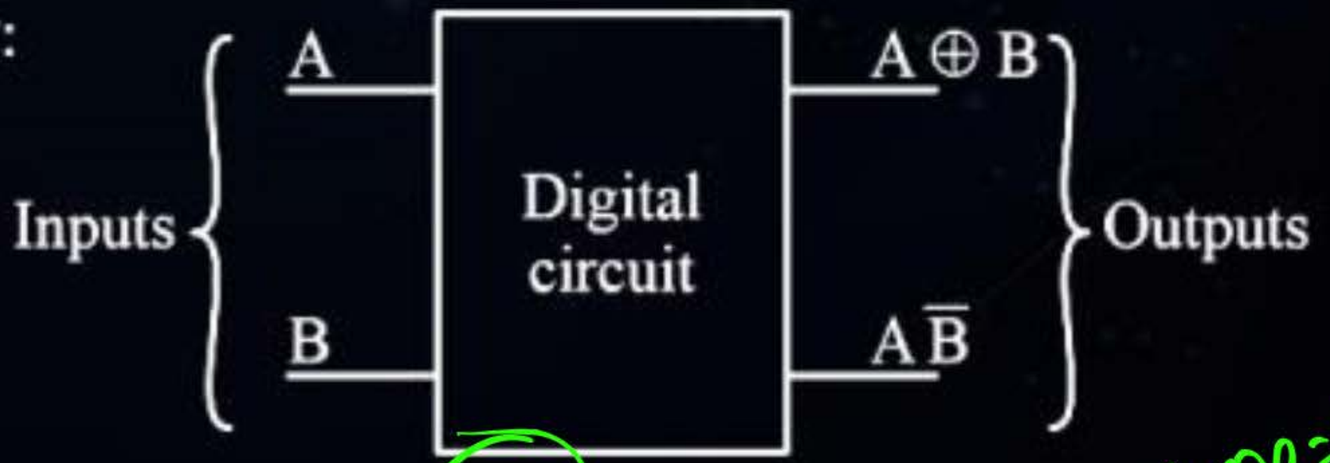
Question **(NAT)**

We have to implement half adder as well as half subtractor. The minimum number of 2-i/P NAND gate required is 6.



Question **(MCQ)**

A digital circuit is as given below:



This circuit implements:

A A + B

B A – B

C B – A

D None of these

B-A

A ⊕ B = Diff

A

B

HS

$\bar{A}B$

A-B

Question **(MCQ)**

A logical circuit is as given below:

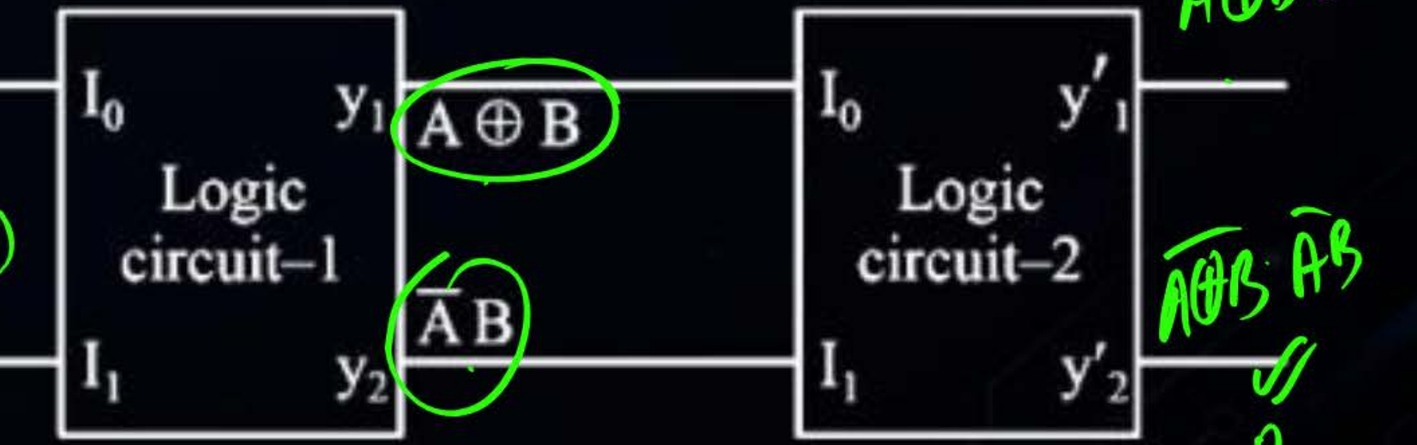
Function of logic circuit – 1 & logic circuit – 2 is same. The output y'_1 will be

(A) $A \oplus B$

(B) $A\bar{B}$

(C) $\bar{A}B$

(D) $A - B$



Question **(NAT)**

n=4 Parallel adder

We have two numbers A and B, and both are 4-bit numbers. To perform complete addition of these two numbers, total number of H.A.s + OR gate required is ___10___.

$(2n-1)$ HA + $(n-1)$ OR

$(8-1)$ HA + 3 OR

7 HA + 3 OR

7 + 3 = 10

Question (MCQ)

x, y, z are i/P$_s$ of a full adder then, sum output of full adder will be '1' if

HW

A $\bar{x}\bar{y} + \bar{y}\bar{z} + \bar{z}\bar{x} = 1$

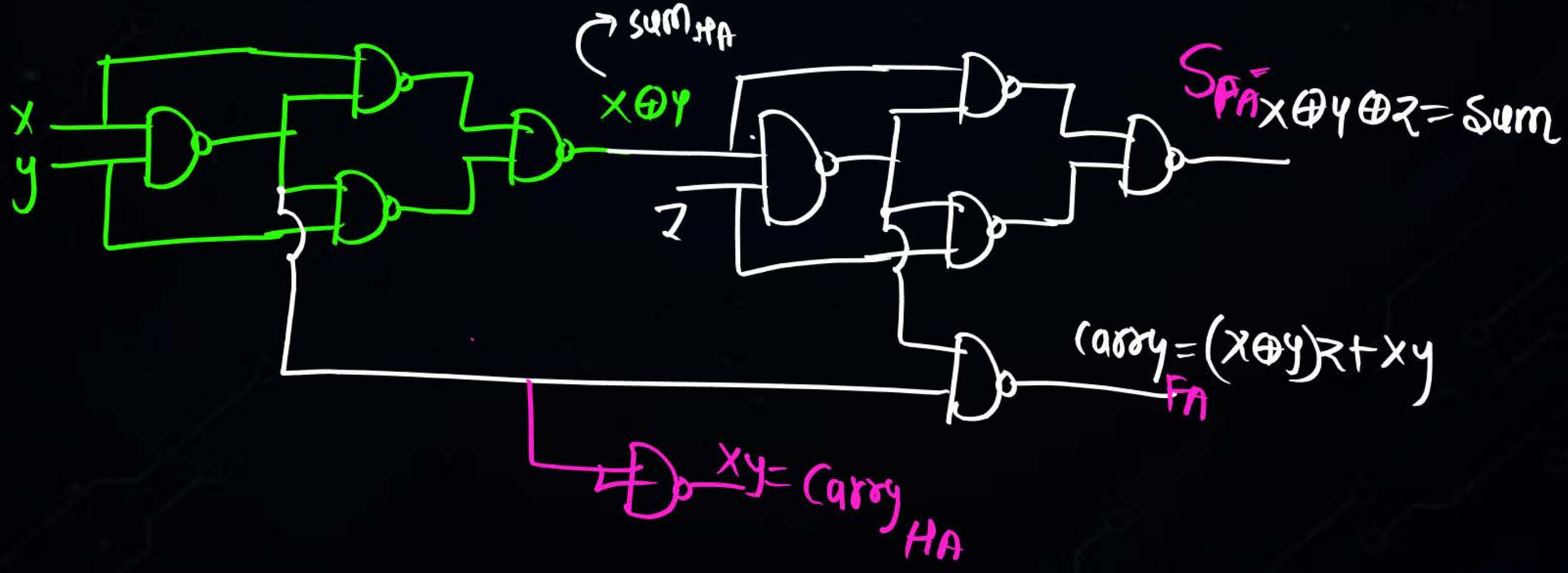
B $xyz = 1$

C $(x + y)(y + z)(z + x) = 0$

D None of these

Question **(NAT)**

We have to implement H.A circuit & full adder circuit together. I/P lines for H.A. is x, y and i/P lines for full adder is x, y, z then minimum number of 2 – i/P NAND gate required is 10.



Question **(NAT)**

We have a 4-bit adder in which each full adder has sum delay of $t_s = 4$ nsec and carry delay of 2 nsec.

In one minute number of additions completed by this full adder is _____ $\times 10^9$.

Question **(NAT)**

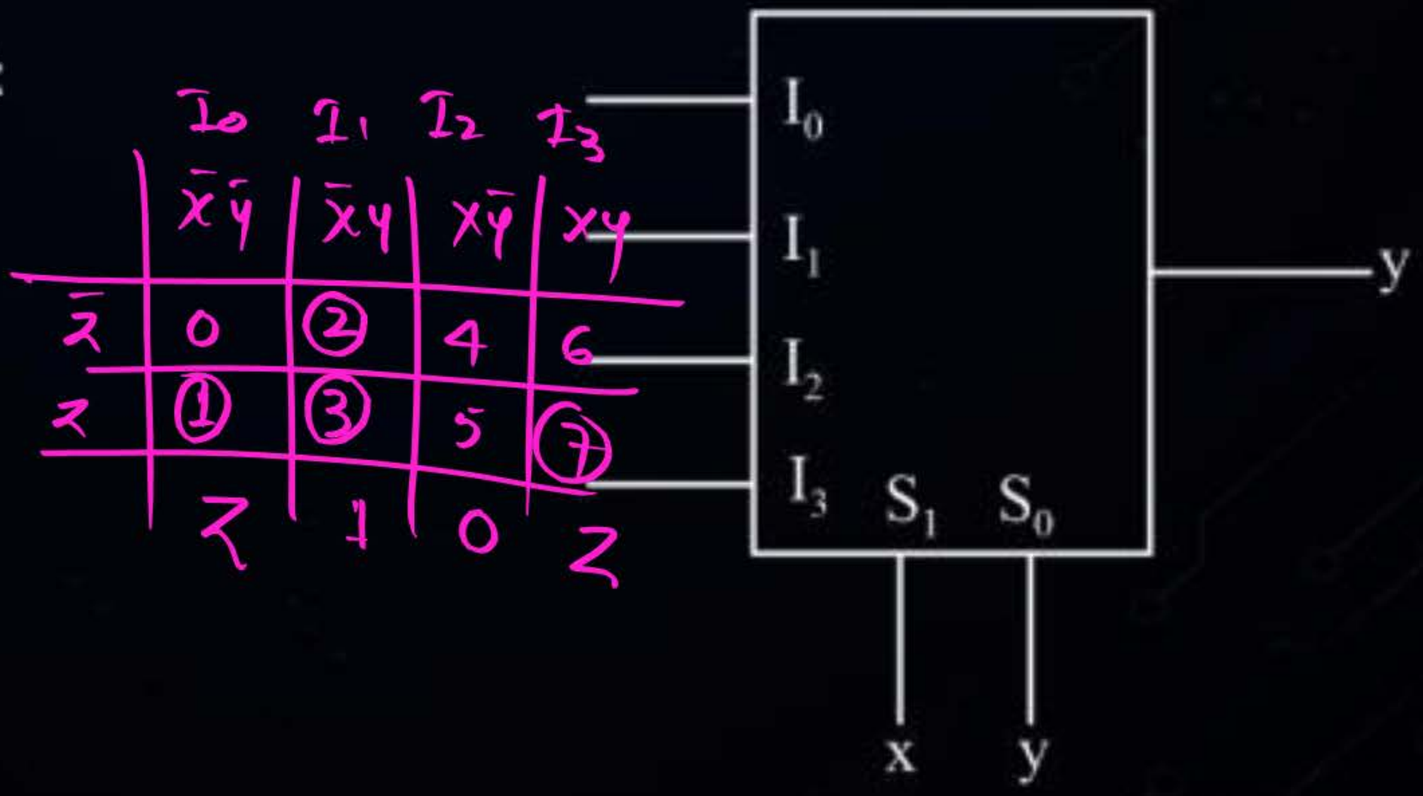
In a 4-bit parallel adder, sum delay and carry delay of each full adder is $t_s = 1$ nsec and $t_c = 5$ nsec respectively. No. of additions performed per sec by adder is N_1. If delays are interchanged then no. of additions performed per sec is N_2, then N_2/N_1 _____.

Question **(MCQ)**

Borrow = $\Sigma m(1,2,3,7)$

In a full subtractor the corresponding input bits of two numbers are x & y [x – y] and input borrow is z and output borrow is B. Output borrow is implemented as shown.

The values of I_0, I_1, I_2 & I_3 are:



A $I_0 = z, I_1 = 1, I_2 = 0, I_3 = z$

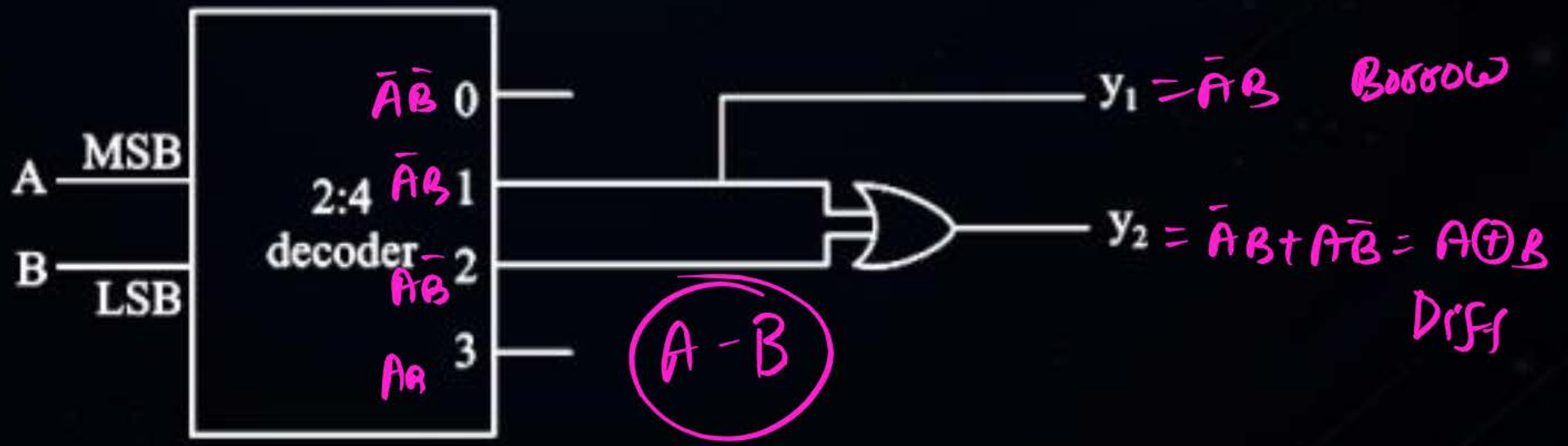
B $I_0 = z, I_1 = 0, I_2 = 1, I_3 = \bar{z}$

C $I_0 = 1, I_1 = z, I_2 = 0, I_3 = z$

D $I_0 = \bar{z}, I_1 = 1, I_2 = 0, I_3 = z$

Question **(MCQ)**

A combinational circuit using decoder and OR gate is as given below:
output y_1 & y_2 combinedly represents



A Half adder

B Half subtractor with subtraction (B – A)

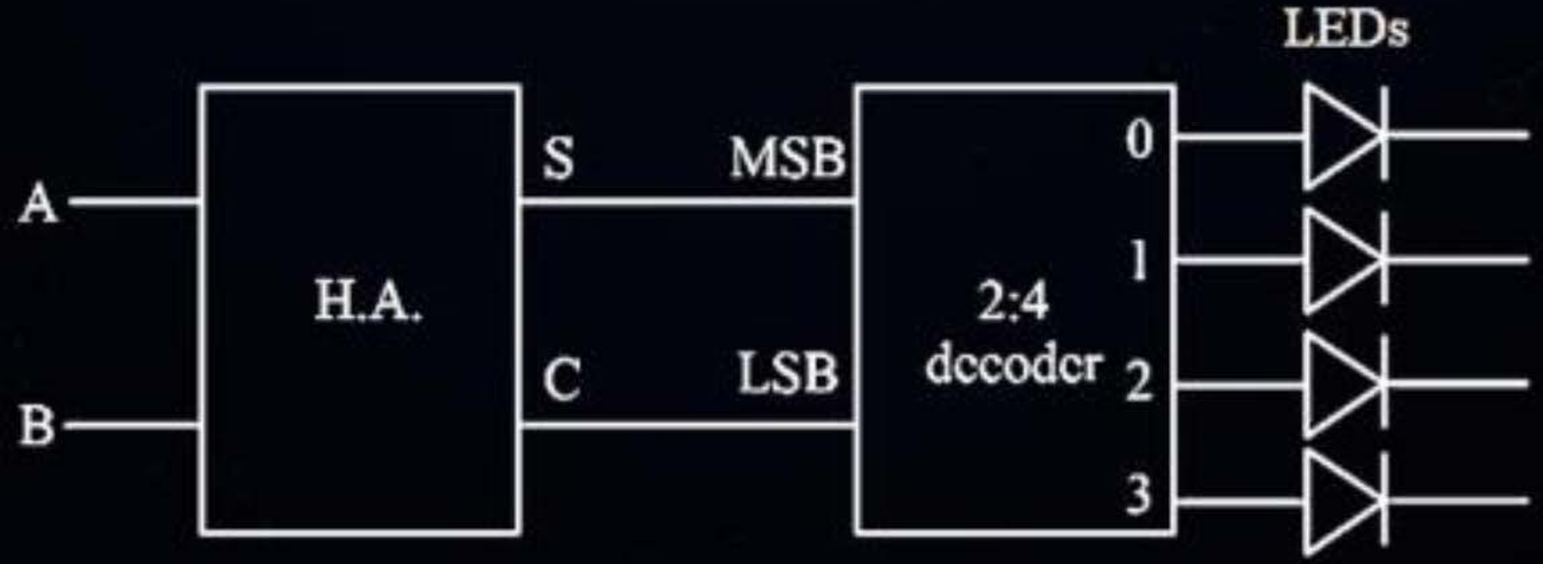
C Half subtractor with subtraction (A – B)

D Comparator o/p with (A > B)

Question **(NAT)**

A combinational circuit is as given below.

When input is A = 0 & B = 1, then LED no. that glows will be_____.



Question **(MCQ)**

Eight 1-bit full adders are cascaded. Each 1-bit full adder generates carry out bit in 10 ns and sum bit in 30 ns. The number of addition performed per second, will be

A 10^9

B 0.5×10^7

C 10^7

D 0.125×10^7

Question **(MCQ)**

#Q. A 16-bit parallel adder is designed using 16, 1-bit full adders. Each 1-bit full adder is designed using X-OR, AND and OR gates. The delay contributed by X-OR gate is 20 ns while that contributed by AND/OR gate is 10 ns. The number of additions per second that a 16-bit parallel adder can perform reliably, will be

A 2.94×10^6

B 3.125×10^6

C 3.33×10^6

D 4×10^6

Question **(MCQ)**

#Q. Each full adder in a 4-bit ripple carry adder, is implemented using one 3-input XOR gate, three 2-input AND gates, and one 3-input OR gate. Let the delay of each 3-input XOR gate be 6ns, that of each 2-input AND gate be 2ns, and that of each 3-input OR gate be 2ns. The total time (in ns) taken by the 4-bit ripple carry adder to do a successful addition operation is _____.

Question **(NAT)**

For implementing full adder and full subtractor we require a decoder of the order n : M, where n is no. of input lines & M is no. of output lines, then value of M is _____.

Question **(NAT)**

To add two 5-bit numbers, we required m H.A.s and n-OR gates then minimum value of (m + n) is _____.

$m = HA = 9$

$n = OR = 4$

$m + n = 9 + 4 = 13$ Ans

Q Design a Circuit which convert

4 bit Binary into Gray code

Thank you DOSTO